UNDERSTANDING AND EVALUATING

EFFECTIVE TECHNOLOGY INTEGRATION IN THE CLASSROOM

by

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A Senior Honors Thesis Submitted to the Faculty of
The University of Utah
In Partial Fulfillment of the Requirements for the

Honors Degree in Bachelor of Science

In

Urban Institute of Teacher Education

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April 2015

ABSTRACT

The current study analyzed four major research questions: (1) Are the attitudes and beliefs of pre-service teachers correlated with constructivist/traditionalist approaches to technology integration? (2) Does prior technology usage predict constructivist/traditionalist approaches to technology integration, as well as pre-service teachers' general attitudes/beliefs about technology? (3) Does taking a technology integration course predict pre-service teachers' confidence in and approaches to technology-based pedagogy? (4) What forms of evaluation materials (typical checklist, abstract checklist, or concrete checklist) facilitate analysis and application of digital materials for effective classroom instruction? Pre-service teachers from the University of Utah (N = 30) indicated their attitudes/beliefs towards technology, rated hypothetical teaching scenarios for effective technology use, and used one of the three checklist forms to evaluate online lesson plans. Results demonstrated a positive, significant correlation between pre-service teachers' general values about technology and traditionalist (teacher-centered) practices but failed to show any significant correlations between prior technology experience and student-centered technology pedagogical practices. Results showed that taking a technology-integration course increased confidence in and intentions to use technology in future classrooms. Finally, participants who used the concrete checklists that contained both principles and cognitive examples of technology integration were more accurate in evaluating the quality of technology integration found in online lessons. Overall, this study suggests that expectations of

technology use in classrooms has resulted in all teachers being more likely to use technology for traditional instructional methods, but more work and training is needed to help pre-service teachers move beyond teacher-centered instruction with technology. Scaffolding technology-based lesson plans with evaluative checklists that include realistic examples of student cognition for principles of technology integration can help teachers analyze technology-based lessons more effectively. However, additional research is necessary to understand how different training and support can help pre-service teachers apply and transfer their understanding of technology integration in a variety of contexts.

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Literature Review

Future educators face tremendous challenges in developing instruction that will prepare students to join a modern workforce in which the use of technology is ubiquitous. A high-quality, 21st century education mandates the inclusion of technology to master skills taught in content areas, teach expression through digital media, communicate and collaborate electronically, discover new information or expand on existing ideas, analyze reliability of resources, gain fluency in computer applications and software, develop problem solving and troubleshooting abilities, and present information to an audience (Kim, Kim, Lee, Spector, & DeMeester, 2013). Although teachers recognize the value of technology integration in the classroom, creating differentiated technology-related learning experiences is a time consuming challenge for most educators (Project Tomorrow, 2012). However, when successfully implemented, technology-based instruction is a powerful pedagogical tool to enhance student learning outcomes; teachers readily agree that digital materials and interactions are an important aspect of instruction in modern classrooms (Hanson & Carlson, 2005).

Teacher Attitudes and Beliefs

A teacher's attitudes and beliefs towards student cognition and effective teaching practices influences much of a teacher's curriculum planning and day-to-day instructional approaches in a classroom environment. As might be expected, a teacher's attitudes and beliefs concerning technology are key determinants in their utilization of technology in instruction. Research has found that a teacher's beliefs and attitudes towards technology use in the classroom influence the quantity and quality of their technology integration (Anderson & Maninger, 2007; Anderson, Groulx, & Maninger, 2011; Ma, Anderson, & Streith, 2005; Ottenbreit-Leftwich, Glazewski, Newby, & Ertmer, 2010). Examples of teacher attitudes and beliefs that influence technology use and integration include self-efficacy beliefs for technology-related tasks, outcome expectations for integrating technology, and value placed on the potential of technology to support student learning (Niederhauser & Perkmen, 2007). All of these factors represent a teacher's personal ideas about technology's value within the instructional environment.

It is somewhat surprising that attitudes and beliefs have been recognized to play such a central role in technology integration, as one might expect that teachers' knowledge about technology and personal expertise with technology tools would be the biggest determinant of whether or not teachers integrate technology into their classrooms. However, teachers' attitudes towards technology have been identified as the most important factor accounting for instructional technology-related decisions in the classroom, even when technology knowledge and expertise are considered (Palak & Walls, 2009). Through both qualitative and quantitative research designs, Palak & Walls (2009) surveyed 133 teachers working in technology-rich schools to determine how teachers' beliefs related to instructional technology practices. Then, guided by these responses, these researchers purposefully investigated two pairs of cases with maximal differences in teachers' beliefs; these four teachers were extensively interviewed and their submitted lesson plans with accompanying reflections about the value of educational technology in the classroom were reviewed. Results from both phases of the study clearly indicate teacher attitudes towards technology as the most significant factor for technology integration, taking precedence over teachers' prior experience and knowledge of technology-related tools (Palak & Walls, 2009).

Teachers' beliefs may be stronger predictors than teachers' technology-based knowledge due to the highly personal quality of beliefs (Ertmer, 2005). Generally speaking, because beliefs are grounded in individual experiences unique to a person's life circumstances, changing one's beliefs may be a daunting task, especially when these beliefs are associated with a profound conviction. Indeed, research with teachers and technology integration is consistent with this interpretation. In an online survey of 217 pre-service teachers' technology-related abilities, beliefs, and intentions, Anderson et al. (2011) found that value beliefs (i.e., the beliefs about the value of a tool to aid in reaching desired student learning outcomes) were the best predictor of pre-service teachers' intentions to use software in their future classrooms as well as frequency with which they intended to integrate technology into their future classrooms. Similarly, qualitative research examining interviews, observations, and electronic portfolios of eight award-winning K-12 technology and general classroom teachers revealed that teacher value beliefs played a critical role in technology integration. Specifically, these researchers found that "the value beliefs that influence teachers' decisions to use technology were motivated by a desire to improve as a professional, ultimately impacting student learning" (Ottenbreit-Leftwich et al., 2010, p. 1330). Although it is clear that there are a plethora of other factors that also influence teachers' intentions and practices, namely an understanding of *how* to implementing technology-related tools, it is important to highlight that research is consistent in the position that "teachers' attitudes toward technology are the most significant predictor for teacher and student technology use" (p. 436, Palak & Walls, 2009).

Even when a teacher values technology and has positive beliefs about its role in modern instructional practice, it should be noted that technology can be used in a variety of ways within the classroom. Just as instruction without technology can be implemented in a variety of ways – from transmission-based lecture to hands-on inquiry – technology-based instruction can be implemented in ways that vary the degree of responsibility and generation required by the students. Decisions about how technology is utilized for instruction may be grounded in the pedagogical beliefs of the teacher that influence their approaches to integrated technology into their classroom practices.

Pedagogical beliefs related to technology. Teachers' pedagogical beliefs refer to a specific type of belief that teachers hold pertaining to the nature of teaching and how classroom instruction should be implemented (Chai, 2010). Pedagogical beliefs are most commonly analyzed using the contrasting dichotomy of constructivist versus traditionalist approaches. Constructivist practices, also known as student-centered practices, include instructional tasking and activities that focus on the student as the main creator of learning materials and resulting knowledge. From a constructivist perspective, the teacher's role is to facilitate learning opportunities by creating an active environment of curiosity, inquiry, and collaboration, serving as a guide and mentor to help the student engage in meaningful forms of generation (Anderson, Groulx, & Maninger, 2011).

Constructivist activities typically include students developing, generating, or creating a product as part of the instructional process (Butcher et al., 2014).

In contrast to constructivist approaches, traditionalist practices emphasize knowledge transmission from the teacher to the learner. The teacher is the source of information that is provided to the learner by activities, examples, and discussion led by the teacher. For this reason, traditionalist practices also can be referred to as teacher-centered instruction. The student's role in traditionalist approaches to teaching often are quite passive; students receive information without being required to integrate, transform, or apply it (Chai, 2010). Accordingly, traditionalist instruction often is associated with lecture-based lessons in which students work individually to encode and remember information.

Different pedagogical beliefs are connected to different types of instruction within the domain of technology use in the classroom. In a survey of 1,230 pre-service teachers in Singapore, Chai (2010) affirmed that constructivist pedagogical beliefs related positively to constructivist use of ICT; that is, teachers' pedagogical beliefs had significant influences on their use of information and communication technology—traditionalist approaches promoted the use of technology for knowledge acquisition while constructivist teaching encouraged the use of collaboration, problem-solving, and communication to spark learning. Kim et al. (2013) similarly found pre-service teachers' pedagogical beliefs to be a strong influencer and predictor of technology integration in future classrooms.

Consistent with national educational reform efforts surrounding generative and inquiry-based practices in science education (National Research Council, 2007, 2011, 2012; National Science Board, 2010), constructivist approaches are the preferred approach to using technology to support meaningful learning. Indeed, research has found that constructivist approaches to teaching are associated with effective, high-level uses of technology in the classroom (Chai, 2010; Hermans, Tondeur, and Vackle, 2008; Kim et al., 2013). In addition, teachers with relatively high constructivist beliefs tend to show higher instances of effective educational technology use (Tondeur, Hermans, van Braak, & Vackle, 2008). Tondeur et al.'s (2008) findings suggest that meaningful professional development for in-service teachers and educational programs for pre-service teachers should address connections between constructivist pedagogy and effective technology integration.

Barriers to Technology Integration

Even with added 21st century demands for technology skills and a general understanding of the value of technology in education, research within the last decade has found that many teachers still do not incorporate technology into their classrooms in effective ways (Culp, Honey, & Mandinach, 2005). Thus, it is important to consider and acknowledge the barriers that teachers face in technology integration. Ertmer (2005) noted that the numerous barriers to technology integration faced by teachers can be categorized as either external or internal factors.

External barriers are defined as those that are not exclusively reliant on the teacher; these barriers include lack of access to equipment/training and limited support from administration. In recent years, external barriers have been greatly diminished as schools have developed significant technology infrastructure (Ertmer, 2005). Ertmer (2005) additionally noted that many school districts have made these structural changes in light of research that has highlighted the overwhelming benefits of effective technology integration in classrooms, including increased student engagement, learning, and efficiency of classroom tasks. Large-scale surveys of multiple stakeholders in education (administrators, teachers, students, and parents) have found that individuals in all roles consider technology access and use to be a critical skill for 21st century learners (Project Tomorrow, 2012); thus, schools and administrators are increasingly willing to fund and obtain technology. Although not all external barriers to technology integration have been resolved, current educational contexts have greatly reduced these obstacles for teachers.

In contrast to external barriers, internal barriers are internal to the teacher and are individual in nature. These include general beliefs, attitudes, and values discussed earlier in this paper: the beliefs and attitudes that teachers hold about the process of teaching as well as specific beliefs and values about the role that technology plays in overall student learning (Ertmer, 2005). Thus, positive beliefs and values about technology and pedagogical beliefs can play an enhancing role in technology integration but negative beliefs and values can serve as a significant barrier to teachers.

Because external barriers have been noticeably reduced (though not eliminated) in recent years, researchers argue that pre-service teacher education and in-service teacher professional development should focus on influencing internal barriers (Ertmer & Ottenbreit-Leftwich, 2013). In support of this effort, Ertmer, Otenbreit-Leftwich, Sadik, Sendurur, & Sendurur (2012) studied 12-award winning technology-integrated teachers and found that the most effective way to convince teachers to use technology is *not* by eliminating external barriers (though that step is still important), but rather by increasing internal factors of knowledge and skills, which have the potential to radically alter existing teacher attitudes and beliefs.

With these obstacles and factors influencing technology integration in consideration, research suggests that it is possible to influence how teachers think about and ultimately use technology. Pre-service teachers have shown change towards more student-centered learning beliefs regardless of prior pedagogical beliefs when taking a technology integration course in which they were asked to reflect on their own beliefs (Anderson & Maninger, 2007; Bai & Ertmer, 2008; Funkhouser & Mouza, 2013). These results illustrate how addressing teachers' attitudes and beliefs may have an impact on pre-service teachers' intentions to effectively use technology in their future classrooms. However, not all forms of professional development result in meaningful changes in teachers' approaches to technology integration. As discussed below, many training and education efforts have failed to achieve desired outcomes.

Efforts to Educate Teachers on Effective Technology Use

In order to increase in-service teachers' knowledge about the merits of and approaches to constructivist technology integration, some school districts have instituted professional development and workshop trainings to educate teachers on effective technology use and to (hopefully) diminish internal barriers to technology integration. Despite these good intentions, these sessions often have not been as effective as intended. Even after training, many teachers continue to use traditionalist technology integration methods, namely a focus on developing skills and knowledge of facts (Mitchem, Wells, & Wells, 2003). While the frequency of technology use in the classroom has been found to increase following professional development training that targets technology integration, the approach that teachers took as they integrated technology often resembled traditionalist, low-level uses (Mitchem et al., 2003). In an examination of the impact of professional development programs aimed at facilitating technology-based instruction, Mitchem et al. (2003) found that, after training, teachers significantly increased their use of technology; however, their data indicates that teachers did so without changing the basic components of the earlier lessons. The result was that teachers still tended to focus on development of skills using traditionalist pedagogy. For example, after a training similar to the one discussed above, some teachers chose to integrate technology through a PowerPoint lecture with multimedia maps to illustrate a topic – for example, the location of important battles in the Civil War. While this example shows an increase in technology use, it aligns with traditionalist pedagogical approaches in which the teacher transmits information and students passively receive content. A higher-level, constructivist approach in this situation could be achieved by requiring students to use technology to find, analyze, and discuss online maps that best communicate the information. Or, to select a digital map from among a series pre-selected by the teacher, annotating the map with important locations and events, and using online tools to explore and explain how the area's topography may have influence battle locations and outcomes. In the constructivist examples, students are engaged in deeper thinking processes (e.g., analysis and evaluation), illustrating an instance in which technology is used in student-centered and generative ways. Thus, teachers must not only understand the content knowledge that they want students to develop, but must understand how the specific capabilities of technology can be used to engage students in content-specific learning and to enhance deep learning processes.

The multiple forms of knowledge that teachers must possess for effective technology integration are described by the concept of Technological Pedagogical Content Knowledge, commonly referred to as TPCK (Koehler, Mishra, & Yahya, 2007). TPCK refers to the relationship between technology, pedagogy, and content knowledge—a combination of intersecting knowledge that teachers must develop in order to effectively select and implement meaningful technology-based instruction. Therefore, in preparing both in-service and pre-service teachers for use of technology in teaching careers, it is essential to develop an understanding of effective technology use as it is related to student cognition and overall knowledge in the content domain. However, professional training often focuses on technical skill development more than implementation support, meaning that educators may learn more about how to interact with a particular technology or digital resource without focusing on how it can transform classroom practices. After participating in a training program on designing computer-based instruction, Valanides & Angeli (2008) found that 40% of the trained teachers failed to use technology as a cognitive tool to enhance learning; instead, these teachers utilized technology as a delivery vehicle to present pictures or textbook content. Thus, teachers may readily adopt technology as a new method to support traditionalist pedagogy, but likely need different forms of support in order to go beyond a traditionalist approach. There is a critical need for materials that help teachers avoid superficial uses of technology and instead focus on effectively selecting and implementing technology in constructivist (student-centered) ways.

Considering that technology-based learning and working will be core requirements for 21st century learners throughout their lives, it is a problem that “although many teachers are using technology for numerous low-level tasks (word processing, Internet research), higher level uses are still very much in the minority” (Ertmer, 2005, p. 26). One issue may be that typical professional development for technology integration fails to distinguish between high- and low-level cognitive processes associated with student use of technology, even when integration methods seem to be student-centered. For example, teachers may be provided with general principles such as to use technology for “active student learning” (Mitchem et al., 2003) without considering whether or not the student activity is behavioral or cognitive (Chi, 2009). For example, students may be working to drag and drop vocabulary terms to appropriate definitions in an online game. Students could be described as learning via active engagement, but their activity does not require knowledge transformation or application. Thus, students are (behaviorally) active but not (cognitively) generative. Closer attention to what is known about high- and low-level cognitive processes during learning may be a key strategy to helping teachers integrate technology in meaningful, constructivist ways within classroom contexts.

Cognitive Processes and Technology Integration

In order to determine whether students are engaged in effective uses of technology, it is necessary to evaluate the cognitive processes which students use while interacting with technology. This evaluation is facilitated by a well-known comprehension framework for student learning known as the Construction-Integration (CI) Model (Kinstch, 1998). The CI Model acts as an outline to identify the levels at which knowledge is acquired during a learning opportunity. These levels – *the surface level*, *the textbase*, and *the situation model* – "characterize the degree to which knowledge is encoded for recall and recognition versus the degree to which knowledge is constructed and integrated for meaningful application" (Butcher, Leary, Foster, & Devaul, 2014, p. 7). Understanding the level at which knowledge is acquired during learning predicts the depth of student understanding as well as the length of time that the knowledge will be retained.

The lowest level of text representation is identified as the surface level. Operating on the surface level means one is able to identify or verify exact passages, sentences, or words; this superficial representation does not demand any understanding of the text (Butcher & Kintsch, 2012). An example of a task that requires surface level recognition is memorizing a poem—the memorizer does not necessarily need to comprehend the meaning of the poem, but simply needs to remember the exact sequence of the words as they appear in the poem (Kintsch, 1998). However, most cognitive tasks do not function on this level because individuals rarely try to memorize texts verbatim. When students say they are trying to memorize a set of learning materials, they usually are trying to create what is referred to as a textbase representation.

A textbase representation is characterized by the encoding of information to understand a text's main ideas or propositions (Butcher & Kintsch, 2012). Memory is required for both surface level and textbase representations; after forming a textbase representation, individuals can recall, recognize, and reproduce information. Individuals with a strong textbase typically can paraphrase learning materials and recognize key definitions, ideas, and facts. For many learners, this is an effective knowledge representation because traditional assessments (fill-in-the-blank items, multiple choice questions) often ask the learner to recall or recognize information that was given in a set of learning materials. However, it is important to note that textbase representations do not represent ideal understanding, as knowledge at this level does not transfer to new contexts and fades quickly (Butcher & Kintsch, 2012).

Of the three levels in the CI Model, the highest level – the situation model – represents the deepest level of understanding. The situation model is formed when incoming information is integrated with prior knowledge, thus transforming the learner's cognitive representation of the information and forming a flexible representation. The situation model represents true learning from a text because this level of knowledge transfers to contexts, domains, and situations different from the original learning context. Learners are able to use situation models broadly, applying their knowledge to new topics and out-of-school environments. In addition, the situation model is the desired level of comprehension because it is a more flexible knowledge representation that is the longest lasting of the three CI Model levels (Butcher & Kintsch, 2012). The knowledge integration necessary for forming the situation model is achieved when students engage in higher-level cognitive processes, including inference, application, analysis, explanation, evaluation, and prediction (Butcher & Kintsch, 2012). These higher-level cognitive processes are associated with constructivist practices, where students are actively engaged in generating, analyzing, and synthesizing information (Butcher et al., 2014). This is another reason why constructivist approaches to technology are critical for effective teachers; these approaches engage students in the types of cognitive processes necessary to form robust, long-term understanding.

Specific to technology classroom contexts, the CI Model may be used as a framework to judge the effectiveness of technology-related instruction. If a technology-related task engages students in the cognitive processes associated with situation model development (e.g., application, analysis, explanation, and analysis), then it should be considered a higher-level use of technology. If the task only requires students to memorize or recall specific information (processes consistent with textbase development), the technology-related task can be classified as a less optimal form of technology integration. While the CI Model has been vigorously studied in the past in the context of comprehension processes and outcomes in laboratory settings, it is now time to study its application in educational practices. In particular, this research explores the value of the CI Model in providing a set of criteria upon which to assess examples of technology integration for new generations of upcoming teachers. The model provides a framework that may help teachers (particularly pre-service teachers) gain insight into the cognitive processes that are needed for effective uses of technology (i.e., higher-order thinking skills associated with Kintsch's (1988) situation model). As digital natives, pre-service teachers may have extensive exposure to technology but are less likely to have previous experience in examining the cognitive processes involved in different types of technology use.

Understanding Technology Integration for Digital Natives

It is important to note that the fast-paced nature of technology adoption in individuals' personal lives may mean that some findings about teachers' beliefs and values about technology integration may be outdated. The future generation of teachers are an unprecedented group of educators due to their designation as "digital natives." In contrast to past generations of teachers, today's upcoming teachers, as members of the Millennial Generation (classified as born between the 1980s and early 2000s), have been identified as more confident and open to change (Pew Research Center, 2010). Evidence from an extensive survey and study from the Pew Research Center from 2006-2010 indicates this new Millennial Generation as highly educated, self-confident, technologically savvy, and ambitious individuals. Specific to technology use, members of the Millennial Generation have been coined as "digital natives" because they are extremely "connected," primarily through social media avenues. For example, Pew Research Center (2010) found that Millennials used social media and text messaging on their cell phones significantly more than older generations. Unsurprisingly, they found that Millennials were also more likely to view technology in a positive light. As a result, pre-service teacher education programs may need to reflect this growing change in attitude towards technology by inventing new methods to support teacher candidates who have more experience, confidence, and understanding of technology. Technology integration classes in teacher education programs have been considered to be the primary method by which pre-service teachers can gain knowledge of technology-related instructional practices (Bai & Ertmer, 2008), but it is important to consider the role of such courses on Millennials' approaches to using technology for instruction. As attitudes and values change and technology becomes a typical component of modern life, technology-integration courses may have less impact than in the past.

However, an argument also could be made for the increasing importance of technology-integration courses for Millennials who are pre-service teachers. The abundance of new technology tools and services available has created an endless and ever-changing set of choices for teachers; it can be overwhelming for many teachers to determine the best technology application or tool to use for a specific educational purpose. For example, the advent of the tablet computer, most commonly in the form of an iPad in educational settings, has drastically changed the landscape of instructional mobility (Project Tomorrow, 2012). Teachers now have access to portable wireless internet on handheld devices as well as a wide variety of apps to meet different educational needs. Although new teachers may be used to selecting and using apps for personal reasons (e.g., social media contributions), it may be less clear how modern technologies can be used to improve and enhance instructional practice in the classroom. Thus, there may be a strong need to educate teachers on appropriate ways to effectively match technology to learning objectives.

Research Purpose

Overall, it is unclear if the relationships between beliefs and intended technology integration will hold for the current generation of pre-service teachers. It may be the case that technology knowledge is more important than beliefs and values, since technology is far more accepted as a necessary part of the personal and academic lives of pre-service teachers today. This study explores the beliefs, attitudes, technology experience, and intentions to use technology in a sample of Millennial pre-service teachers. It also examines whether a technology-integration course during pre-service training has a significant impact on Millennial's intentions to use technology in the classroom. Finally, this study also seeks to determine whether scalable forms of support that target cognitive processes can help pre-service teachers recognize constructivist versus traditionalist methods of technology integration.

Research Questions

Aligned with the research purpose, this study target four major research questions: (1) Are the attitudes and beliefs of pre-service teachers correlated with constructivist/traditionalist approaches to technology integration? (2) Does prior technology usage predict constructivist/traditionalist approaches to technology integration, as well as their general attitudes/beliefs about technology? (3) Does taking a technology integration course predict confidence in and approaches to technology-based pedagogy? (4) What forms of evaluation materials (typical checklist, abstract checklist, or concrete checklist) facilitate analysis and application of digital materials for effective classroom instruction?

Methods

Participants

Participants were 30 undergraduates enrolled in secondary and elementary education programs at the University of Utah. Participant age ranged from 18-26 (M = 21.5); the sample consisted of 8 males and 21 females. Selection of participants was completed through an online participant pool database; students enrolled in this study to partially fulfill a course requirement for introductory educational psychology classes.

Research Design

This experimental study utilized a between-subjects design. Participants were randomly assigned to one of three experimental conditions (see Table 1).

Table 1.

Experimental Conditions

	Condition		
	Concrete Checklist Principles + Cognitive Examples	**Abstract Checklist** Principles	**Typical Checklist**
Description	Checklist provided high- and low-level principles with cognitive examples to consider for technology integration.	Checklist provided high- and low-level principles for technology integration without cognitive examples.	Checklist provided four simple, general questions to consider effectiveness of technology integration.

Materials

Survey on Attitudes, Beliefs, and Intended Technology Integration. This survey measured participants' attitudes, beliefs, and intentions to use technology in the classroom using questions targeted at participants' previous interaction, experience, and confidence with technology and technology-related administrative tasks. The survey was administered over nine pages using an online survey program (SurveyMonkey) with approximately 10-15 items per page for a total of 122 items. The survey began with 18 demographic questions addressing participants' gender, age, completion of educational technology classes in the pre-service education program at the University of Utah, teaching preferences in future grade level/subject area, and frequency with which participants used social media outlets (e.g., Facebook, Instagram, Twitter, Snapchat, and LinkedIn). Items for the main portion of the survey were selected from previously used and validated instrumentation as described below.

Abilities, Self-Efficacy, Value Beliefs, and Constructivist Beliefs Scale. Drawn from Anderson, Groulx, & Maninger (2011), items from this scale addressed the following factors: confidence in performing technology related tasks, technology self-efficacy beliefs, technology value beliefs, constructivist pedagogical beliefs, and intentions to use technology. 33 items from this scale were utilized in the current study. Participants selected the extent to which they agreed with statements using a 6-point Likert-style scale rating system (i.e., strongly disagree, slightly disagree, disagree, agree, slightly agree, and strongly agree). Example items from this scale include: "I feel confident that I could select appropriate software and use it for a particular task"; "I feel confident that I could plan and implement instruction that integrates technology into my curriculum." Items related to technological skill competence were specific to word processing, spreadsheet, presentation, and

Internet tasks. For example, participants rated their confidence in "inserting graphics into a document" and "creating and using headers and footers" in word processing tasks.

Intrapersonal Technology Integration Scale (ITIS). From Niederhauser & Perkmen (2007), the ITIS portion of the survey was organized according to four subscales related to educational technology use: self-efficacy, outcome expectations, interest, and behavior intentions. In the present study, five items were incorporated to identify participants' confidence in implementing instructional technology. Participants rated their agreement with statements using the same 6-point Likert-style response scale as described for the previous scale. Example items include: "I can teach relevant subject matter with appropriate use of instructional technology"; "I can select appropriate technology instruction based on curriculum standards-based pedagogy." Six items were incorporated from this scale to assess participants' beliefs and attitudes towards the value of technology, as exemplified in the following statements: "Effectively using instructional technology in the classroom will increase my sense of accomplishment"; "Using instructional technology will make my teaching more exciting."

Teacher Attribute Survey (TAS). From Vannatta & Fordham (2004), this survey covers a wide variety of factors that influence teachers' use of technology in the classroom. Of specific interest to this study were the questions regarding pedagogical beliefs; 12 items were used that specifically measured participants' pedagogical beliefs as either constructivist or traditionalist in nature. For example, the extent to which participants agreed with the statement like "I believe in developing my classroom as a community of learners" or "I plan to function in my classroom as a learner and partner in learning with my students" indicated their adherence to constructivist approaches. In contrast, a participant who rated high agreement with statements similar to "It is better when the teacher, not the student, decides what activities are done" or "Teachers know a lot more than students; they shouldn't let students muddle around when they can just explain answers directly" were categorized as aligning with traditionalist beliefs.

Teaching, Learning, and Computing (TLC). The original TLC Survey was created in 1998 but was revised and updated in Kim et al. (2013). The more recent version acted as a source of survey items for the current study. 14 items were utilized to determine teachers' conceptions of teaching along the constructivist versus traditionalist continuum. Participants again rated the extent to which they agreed with statements, such as: "Students won't really learn a subject unless you go over the material in a structured way. It's my job to explain, to show students how to do the work and to assign specific practice" (higher rating of agreement aligns with traditionalist mentality); "The most important part of instruction is that it encourages 'sense-making' or thinking among students. Content is secondary" (higher rating of agreement aligns with constructivist ideology).

Seven additional items related to assessment practices were included to gather more information about participants' pedagogical beliefs. Participants who judged open-ended questions, group projects, and student presentations/portfolios as more useful instruments to assess student learning were characterized as having constructivist beliefs. Participants who chose standardized test results, essay tests, or multiple choice questions (traditional forms of assessment) as most useful were considered to identify with traditionalist pedagogical beliefs.

Eight items from the TLC were also included regarding disadvantages to technology integration. Examples include: “Computers are too unpredictable—they crash or the software doesn’t work right”; “Many students use computers in order to avoid doing more important school work”; “Students can cheat easier—copying work and turning it in as their own.” Agreement with these statements signaled traditionalist attitudes and beliefs towards technology.

The TLC Survey also contained items that measured intentions to use technology by questioning participants’ student objectives and teacher objectives for computer use in the classroom. Examples of student computer use objectives (10 items) included “mastering skills just taught” and “learning to work collaboratively”; agreement with the first statement signals traditionalist beliefs while agreement with the latter statement indicates constructivist beliefs. Teacher objectives (13 items) for computer use were likewise rated according to agreement with statements such as to “prepare multimedia presentations for the classroom” (traditionalist approach to technology integration) and “communicate with colleagues, other professionals, or parents” (constructivist idea of using technology to communicate).

Self-created Technology-Constructivist Belief Items. In the creation of the survey for this study, a gap was identified in the lack of questions specifically regarding the relationship between constructivist approaches and technology-based instruction. The author of this study therefore created 10 items to include in this survey to identify the amount to which participants identify specifically with constructivist-technology practices. High agreement with statements similar to the following show high constructivist-technology beliefs: "Technology is not an end in and of itself, but rather a pedagogical tool used to enhance learning outcomes"; "Technology creates opportunities for students to be more active participants in learning—they have more choice and influence in what and how much they learn"; "Technology is best used when it enables students to do authentic work (e.g., communicating/collaborating online, creating multimedia presentations)."

Checklists. Checklists were created to provide a scaffold for teachers to use when evaluating digital materials for classroom evaluation. The purpose of implementing a checklist was to investigate how an easy-to-use tool that incorporates information about student cognition would influence pre-service teachers' evaluation of approaches to effective technology integration. The checklists were the only physical material used in this study, meaning participants checked items on a paper checklist using a pen; all other components were completed via the SurveyMonkey online interface. The checklists were created according to three conditions (see Table 1) of scaffolded support to determine what level of support best enabled participants to identify high-level, constructivist over low-level, traditionalist uses of educational technology.

Connection to CI model. The checklist items were created by the faculty advisor utilizing Kintsch's (1998) CI model of text comprehension. Because the textbase and situation model levels emphasize the differences between high-level cognitive processes (e.g., inference, application of knowledge) and low-level cognitive process (e.g., memorization, repetition, rehearsal of given information), the checklists included statements addressing both types of student cognition to help participants identify the discrepancy between the two levels of understanding. Of the three checklists, a "typical" checklist was created to represent the type of general advice that commonly is presented to teachers. Its simple prompting and layout resembles the abstract support commonly given to teachers to evaluate technology-related tasks or activities (as an example, see http://www.educatorstechnology.com/2012/12/a-list-of-great-checklists-every.html). The other two checklists, the abstract and concrete, were more explicit and detailed in their focus (as described below).

Typical checklist. The typical checklist (see Appendix A.1) was the most basic form with only four generic prompts provided for participants to use to determine the effectiveness of the technology-related lesson plan. One such question was "Are students actively working with materials/technology/websites?" Participants in this condition simply checked the box to the corresponding statements if they agreed the statement was exemplified in the digital lesson.

Abstract and concrete checklist. These checklists were organized into two sections: generative methods and quality of cognitive processing. In the generative methods section, the statements focused on evaluating whether the technology-based activity was teacher-centered (traditionalist) or student-centered (constructivist). For example, prompts asked participants to evaluate if students were involved in "Creative problem-solving or coming up with new ideas," if the "Teacher manipulates the technology and/or directs interactions with the technology," and "Students find technology content or determine what/how to use digital information." Participants checked the box next to statements that they believed to be true for a given technology-based activity.

The quality of cognitive processing section focused on the contrast between low-level (textbase focused) versus high-level (situation model focused) cognitive processes. Participants were prompted to look for examples of the following low-level cognitive processes in the technology-based activity: memorization, repetition, rehearsal of provided information, recall of given materials, and rote performance of a provided, step-by-step procedure. Participants were also asked to identify whether the technology-based activity required students to engage in high-level cognitive processes such as: application of knowledge, inferences made about given content, integration of new and prior knowledge, evaluation of data or evidence, explanations that go beyond provided materials, and focus on understanding concepts or ideas.

The concrete and abstract checklists were organized in the same way with one notable exception: the inclusion of concrete, realistic examples that illustrated the principle in action. The abstract checklist (see Appendix A.2) listed the principles for technology integration without cognitive examples; the concrete checklist (see Appendix A.3) included the same principles, but also provided an example of the principle in a realistic scenario related to educational technology. For example, the abstract checklist contained the prompt, "Recall of given materials." The concrete checklist augmented this principle with "Example: Students are asked to select three ways to tell if something is alive; all three correct choices were discussed in a previously-viewed video" written below the principle. The purpose of differentiating the abstract and concrete checklists was to determine if providing an explicit example demonstrating how to identify the principles in the checklist would aid teachers in evaluating technology-based activities more than a checklist without examples.

Technology Integration Vignettes. In order to gauge participants' understanding of technology integration, a series of vignettes were created. Vignettes were descriptions of hypothetical lessons involving technology. Participants rated these vignettes along a series of dimensions in order to determine their ability to identify high-level, effective technology use in the classroom. Scenarios varied according to student-centered and teacher-centered approaches to teaching, the quality of integration, and the potential for student learning. Vignettes were chosen as an evaluation tool because the rating of teaching scenarios provided an objective approach to collect reliable and valid data as opposed to solely relying on self-reported information and intentions from the survey.

A total of nine vignettes were included in the study ranging from low-level, traditionalist uses of technology to high constructivist use that targeted higher-level cognitive processes. The vignettes, approximately three paragraphs in length each, include real examples of technology tools and apps for use in the classroom (e.g., Microsoft PowerPoint, Google Doc, etc.). With the aid of the faculty advisor, the researcher ranked each of the nine vignettes on 16 items as low, medium, to high. Of the nine vignettes, three were ranked overall as exhibiting low levels of constructivist technology integration, three as medium, and three as high. The differences in the vignettes was purposefully created to highlight whether or not participants could identify high-level, constructivist uses of technology from medium- to low-level cognitive uses. A vignette was designated as medium if it contained components of both constructivist and traditionalist technology integration (e.g., uses a PowerPoint to present information, then allows students to manipulate an interactive simulation to reinforce ideas taught). For examples of a high, medium, and low vignette teaching scenario, see Appendix B.

Vignette Evaluation Items. Each of the nine vignettes were evaluating using 16 items that targeted the following dimensions of instructional quality: quality of technology integration, quality of digital materials used, opportunities for student learning, and level of student engagement. Examples of evaluation items included: "This teacher selected online or digital materials that go beyond what traditional materials could provide"; "This teacher created opportunities for students to interact with technology in meaningful ways"; "Students would be engaged in deep learning processes during the lesson." Participants selected the extent to which they agreed with the statements using the same 6-point Likert scale used in the survey (i.e., 1 = strong disagree, 6 = strongly agree). For the complete list of vignette evaluation items, see Appendix C.

Online Lesson Plans. The checklists described above were used to evaluate four online lesson plans. Each participant was given four checklists, one per online lesson plan, according to the condition they were randomly assigned to (typical, abstract, or concrete). Of the four lesson plans, two were judged by the researcher (with consultation by the faculty advisor) to represent high-level uses of technology and constructivist pedagogy, such as analysis and creation of new knowledge; the other two lesson plans were considered to be low-cognitive uses of technology that focused on traditionalist approaches to integration, such as repetition and recall of information. The two lesson plans considered to be effective uses of technology were "Mr. Droplet's Journey Home WebQuest" (https://sites.google.com/site/3140ps/introduction) and "What's My Structure?" (http://www.uen.org/Lessonplan/preview?LPid=166). The other two lesson plans, "Ancient Roman Helmets WebQuest" (https://sites.google.com/site/ancientromanhelmets/) and "Sharks!" (http://ia.usu.edu/viewproject.php?project=ia:8292) were judged as ineffective uses due to their focus on fact-based learning, memory, and recall of facts. Participants were provided links to each of the four lesson plans with the instruction to read through the lesson activities and materials then complete the corresponding checklist. Participants' ability to identify the effective lesson plans was identified through the boxes checked to corresponding statements on the provided checklist.

Procedure

The study was designed as a self-guided sequence of questions and instructional videos using the online survey tool SurveyMonkey. The total session took between one to two hours depending on the participant and condition.

Upon arrival at the experiment, participants were taken through an informed consent procedure and assigned randomly to one of the three experimental conditions. Participants were provided with the appropriate checklist for their assigned condition, but were not yet instructed to begin using the checklist. Participants next watched an instructional video that introduced them to the experiment and explained that they would be answering questions about their beliefs about technology integration and evaluating a variety of online resources. The instructional video reiterated the fact that the study was voluntary and the participants could withdraw at any time for any reason. Participants began the study by entering their randomly assigned participant ID number on the computer.

Next, participants watched an instructional video that explained that the first task: completing the online survey. The video noted that there were no right or wrong answers on the survey; participants were instructed to select the answer that most closely corresponded to their personal beliefs and ideas. Survey items were then presented via computer. After completing the survey, participants watched another instructional video that described the vignette task. Participants were told that they would read a hypothetical lesson using technology, then would be asked to rate these vignettes on the quality of technology integration, digital material used, opportunities for student learning, and level of student engagement. The video stated for participants to use their *current* thinking based on their own evaluation to rate the nine vignettes.

After the initial vignette evaluation, participants viewed the next instructional video which instructed them to begin using the provided checklists in order to evaluate four technology-related lesson plans. The video explained that participants needed to click on the link to open each lesson plan, read the procedure and materials, and then use the appropriate checklist to check the boxes if they believed the statement accurately described the activities in the lesson plan. Participants progressed to the subsequent lesson plan by returning back to the study window and clicking "Next" at the bottom of the screen.

In order to gather more qualitative data on participants' ideas and beliefs about technology integration, the next stage of the study involving participants revising a lesson plan about heat transfer. This lesson was purposefully chosen for its mixed use of pedagogical approaches; that is, the lesson used some tasks aligned to constructivist-technology beliefs, but also had some instances of repetitive, traditionalist approaches to teaching with technology. Another instructional video informed participants that their task was to explain how they would revise specific steps of the heat transfer lesson plan. The next three pages of the study included a prompt to revise a specific step of the lesson plan with a textbox available below for participants to type their response.

The final stage of the study was to reevaluate the vignettes to determine if participants' ideology or approach to technology integration had changed as a result of their condition. To test this possibility, the instructional video explicitly reminded participants to use the checklists provided to help them decide on the ratings for each vignette. After completion of the vignettes reevaluation, a final instructional video thanked participants for their time and informed them that they had received credit for their participation. The participants were instructed to leave their checklists at the desk and to click "Done" at the bottom of the screen to indicate completion of the online portion. After participants exited the computer lab at the end of the study, the researcher gathered checklists and organized them according to condition for later analysis.

Results

Technology Beliefs and Attitudes Relationship to Pedagogical Beliefs

Research Question 1. Are the attitudes and beliefs of pre-service teachers correlated with constructivist/traditionalist approaches to technology integration? This question was addressed by analyzing bivariate correlations between survey data responses on items targeting teachers' beliefs and attitudes about technology and their pedagogical beliefs. Appendix D shows how individual items were compiled into categories corresponding to technology attitudes/beliefs and pedagogical beliefs. Correlations between technology attitudes/beliefs and pedagogical beliefs are shaded in Table 2. There were significant, positive correlations between all three categories of technology attitudes and beliefs (see Table 2). Thus, pre-service teachers in this sample who strongly agreed with the value of technology in one area also agreed with its value in the other areas.

Correlations between categories of technology attitudes/beliefs and pedagogical beliefs showed no significant correlations between technology attitudes/beliefs and student-centered practices or assessments. There was a significant, positive correlation ($r = .47$) between pre-service teachers' general agreement with technology as a useful tool to meet instructional objective and their agreement with teacher-centered practices in the classroom. There also were significant, positive correlations between agreement with teacher-centered assessment methods and all three categories of technology beliefs/attitudes (technology valuable in meeting learning objectives, r = .46; technology important to instructional practices, r = .49; technology important to effectiveness as teacher, r = .40).

Table 2.

Bivariate correlations between technology attitude/beliefs and pedagogical beliefs

	1	2	3	4	5	6	7	8
1. Technology valuable in meeting learning objectives	1							
2. Technology important to instructional practices	.78**	1						
3. Technology important to effectiveness as teacher	.81**	.80**	1					
4. Agreement with teacher centered practices	.47**	.32	.34	1				
5. Agreement with student-centered practices	.29	.30	.17	.30	1			
6. Agreement with teacher-centered assessment methods	.46**	.49**	.40*	.43*	.08	1		
7. Agreement with student-centered assessment methods	.23	.31	.30	-.28	.34	.10	1	
8. Agreement with student-centered technology use	.74**	.67**	.67**	.13	.45*	.20	.38*	1

When responding to items that directly assessed agreement with student-centered uses of technology (rather than technology use in general), participants' responses were correlated (significantly and positively) to their agreement with student-centered pedagogical practices (r = .45) and assessments (r = .38) but not to teacher-centered practices (r = .13) or assessments (r = .20). Thus, student-centered pedagogical beliefs do predict agreement with student-centered uses of technology.

Agreement with student-centered uses of technology also was significantly correlated with increased agreement about the value and importance of technology (technology valuable in meeting instructional objectives, $r = .74$; technology important to instructional practices, $r = .67$; technology important to effectiveness as teacher, .67).

Research Question 2. Does prior technology usage predict constructivist/traditionalist approaches to technology integration, as well as their general attitudes/beliefs about technology? Responses about pre-service teachers' confidence and experience in technology-related skills (i.e., word processing, spreadsheets, presentations (PowerPoint), the Internet/World Wide Web, and social media use) were correlated with agreement to student-centered technology use and categories of technology attitudes/beliefs. Table 3 shows the bivariate correlations between these variables.

Overall, results indicate pre-service teachers' confidence in technology skills and experience were not significantly correlated to their agreement with student-centered uses of technology nor to their overall attitudes and beliefs about technology. Correlations showing these relationships are shaded in Table 3. However, self-reported social media use was significantly and positively correlated with agreement that technology is valuable in meeting instructional objectives ($r = .45$) and that technology is important to instructional practices ($r = .46$).

Table 3.
Bivariate correlations between prior technology usage, technology pedagogical approaches, and beliefs and attitudes about technology

	1	2	3	4	5	6	7	8	9
1. Word Processing	1								
2. Spreadsheets	.57**	1							
3. Presentations (PowerPoint)	.81**	.44*	1						
4. Internet/World Wide Web	.53**	.46*	.48**	1					
5. Social Media Usage	.27	-.08	.10	.19	1				
6. Agreement with student-centered technology use	.30	.04	.17	.24	.27	1			
7. Technology valuable in meeting learning objectives	.22	.03	-.01	.33	.45*	.74**	1		
8. Technology important to instructional practices	.23	.01	-.01	.19	.46*	.67**	.78**	1	
9. Technology important to effectiveness as teacher	.18	-.01	.03	.28	.35	.67**	.81**	.80**	1

Interestingly, social media use was the only technology skill that was not positively related to confidence and experience in other forms of technology. Word processing, spreadsheets, presentation (PowerPoint), and Internet/World Wide Web experience were all significantly, positively correlated.

Research Question 3. Does taking a technology integration course predict confidence in and approaches to technology-based pedagogy? Participants confidence was assessed using a multivariate analysis of variance (MANOVA) where completion of a technology integration course (Yes vs. No) was the independent variable and self-reported confidence in teaching technology, ability to integrate technology into their future classrooms, in using the web effectively, and in using technology were the dependent variables. A trend was seen for technology course completion in the multivariate analysis ($F_{(4, 25)}$ = 2.1, p = .10, η_p^2 = .25). Univariate results were examined in order to interpret the multivariate effect.

Univariate results showed an significant main effect of taking an introductory technology integration course on students' self-reported confidence in teaching technology ($F_{(1, 28)}$ = 4.1, p = .05, η_p^2 = .13), confidence in integration technology into their future classrooms ($F_{(1, 28)}$ = 6.1, p = .02, η_p^2 = .18), and confidence in using the web effectively ($F_{(1, 28)}$ = 6.2, p = .02, η_p^2 = .18). There was a trend of taking an introductory technology integration course on students' overall confidence in technology use, although it did not reach the level of statistical significance ($F_{(1, 28)}$ = 3.4, p = .08, η_p^2 = .11). As seen in Table 4, students who took an introductory technology integration course showed higher confidence in all cases.

A one-way analysis of variance (ANOVA) was conducted to determine the effect of taking a technology integration course on students' agreement with student-centered uses of technology. Results demonstrated no difference among students (F < 1); mean agreement is seen in Table 4. Overall, agreement with student-centered technology use was high for both groups, averaging just over 5 on a 6-point scale. Thus, all pre-service teachers tended to agree with the potential utility of these technology approaches.

Table 4.
Means (and standard deviations) in confidence about technology use for students who have and have not taken a technology-integration course

	Confidence in teaching technology	Confidence in integrating technology	Confidence in using the web effectively	Confidence in overall technology use	Agreement with Student Centered Tech Use
Took course	5.4 (.51)	5.4 (.57)	5.5 (.46)	5.3 (.52)	5.1 (.52)
Did not take course	4.9 (.75)	4.7 (.84)	4.9 (.76)	4.7 (.96)	5.1 (.50)

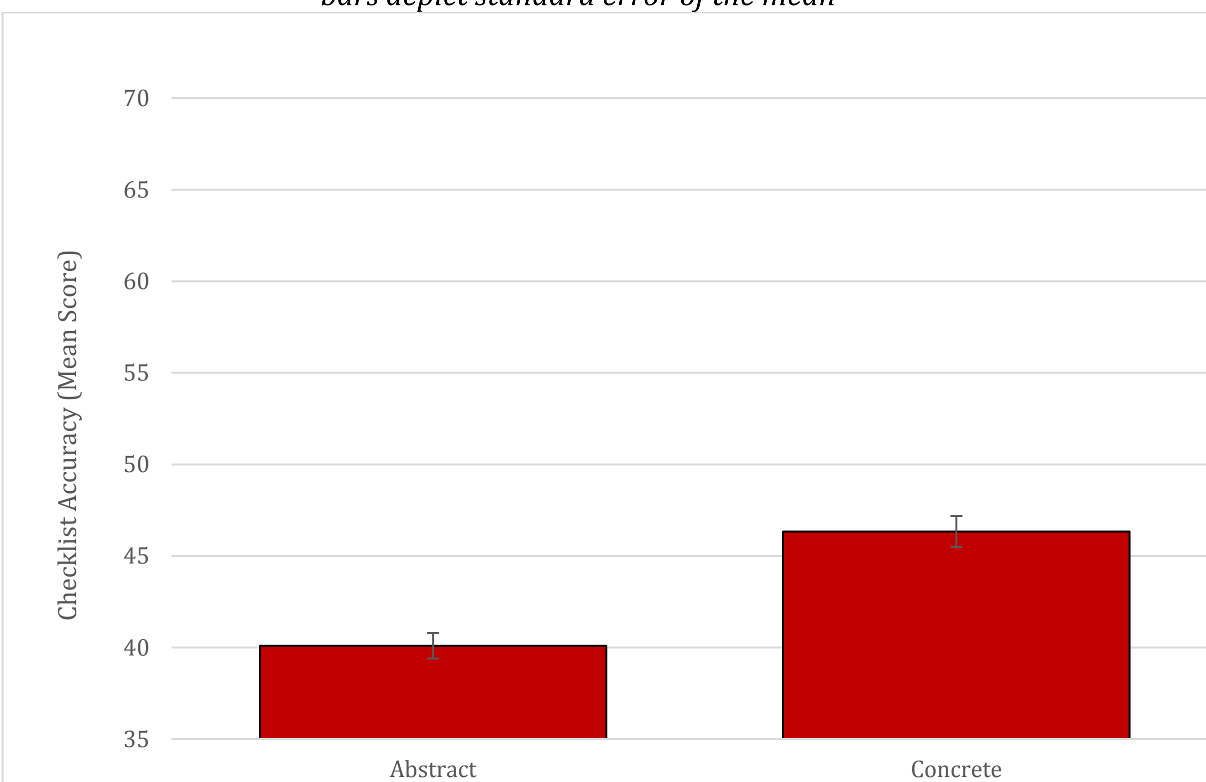
Checklist Condition Effect

Research Question 4. What form of evaluation materials (typical checklist, abstract checklist, or concrete checklist) facilitate analysis and application of digital materials for effective classroom instruction? A RM-ANOVA was conducted to determine if the form of checklist that pre-service teachers used during evaluation impacted their judgments of vignettes that described classroom scenarios in which technology was utilized. Test time (pre- vs. post-test) was the repeated factor, checklist type was the independent variable, and score on vignette ratings was the dependent variable. Results showed no effect of test time ($F_{(1, 24)} = 1.7$, $p = .21$), no effect of condition ($F < 1$), and no interaction between test time and condition ($F < 1$).

In order to determine if participants were able to use the checklist effectively, a one-way ANOVA was conducted to assess the impact of the abstract (principles) and concrete (principles + cognitive examples) checklists on pre-service teachers' evaluations of online lesson plans. The typical checklist was not included in this analysis because it did not include the same number or type of evaluation items as the other checklists. Thus, this analysis determined if the presence of concrete examples helped pre-service teachers analyze online lessons as they used the checklist.

As seen in Figure 1, there was a main effect of condition on pre-service teachers' accuracy in evaluating the online lesson plans ($F_{(1, 17)} = 5.7$, $p = .03$, $\eta_p^2 = .25$). Participants who used the concrete version of the checklist ($M = 46.3$, $SD = 6.5$) were better able to assess the online lesson plans than students who used the abstract checklist ($M = 40.1$, $SD = 4.9$).

Figure 1.
Mean scores on accuracy of lesson plan evaluation by experimental condition. Error bars depict standard error of the mean



Discussion

Relationship between Technology Attitudes and Pedagogical Approaches

The first research question of this study examined the relationship between attitudes and beliefs of pre-service teachers and pedagogical approaches to technology integration. The three attitudes/beliefs categories used (see Appendix D for categorization) were positively and significantly correlated with one another, suggesting participants in this study agreed that there is value in technology to meet a variety of instructional and professional needs. Specifically, participants noted the importance of technology in addressing student learning objectives, teacher administrative objectives, and its ability to make teaching more meaningful, satisfying, and effective. This finding is consistent with the work conducted by Project Tomorrow (2012) in which Millennials showed agreement with the power of technology to enhance aspects of work and life. Pre-service teachers' responses from this study suggest that the upcoming generation of "digital native" teachers similarly see the inherent value of technology to act as a powerful tool in improving educator effectiveness.

Interestingly, correlations between technology attitudes/beliefs and pedagogical beliefs showed no significant correlations to constructivist (student-centered) pedagogy and assessments. Instead, results from this study demonstrated a positive, significant correlation between pre-service teachers' general values about technology and traditionalist (teacher-centered) practices. These results differ from the findings of earlier research, which has found that positive attitudes and beliefs towards technology have been associated with an openness to trying new and alternative instructional methods that are associated with constructivist (student-centered) pedagogy (Anderson et al., 2011). What might be the reason for this discrepancy? One possibility is that the current educational climate emphasizes the use of technology as part of 21st century classrooms. Therefore, all pre-service teachers may be planning technology integration as an expected part of their instructional practices. In years past, technology integration may have been considered a more “cutting-edge” practice, meaning that teachers who were more likely to adopt technology may have been those who also were more likely to try non-traditional instructional practices. Current pre-service teachers may consider technology as a key method to implement traditionalist pedagogy. For example, teachers commonly use presentation software (e.g., PowerPoint) to communicate or introduce a topic of study. This interpretation is consistent with research conducted by Mitchem et al. (2003) in which, even after professional development training, technology-related instruction resembled traditionalist, low-level uses of technology in the classroom. Therefore, the results from the current work align with the idea that while pre-service teachers’ beliefs and attitudes towards technology are generally positive, targeted work and training is needed to help teachers move past traditionalist pedagogy toward constructivist methods of technology integration.

Results from the current research also indicated significant correlations between all three categories of pre-service teachers’ technology attitudes/beliefs and traditionalist assessment practices (e.g., standardized tests, multiple choice quizzes). These results may reflect frequent instructional emphasis on using technology to assess or practice basic skills and knowledge. For example, quiz games for arithmetic or literacy activities that test vocabulary knowledge or phoneme awareness. In addition, with the implementation of digitally-based standardized tests (e.g., SAGE testing in Utah) in which students are required to be familiar with computer-related skills and the testing interface, some teachers may implement "drill-and-skill" technology tools with the intention to help students gain fluency in computer-based skills as needed for standardized testing. These examples of technology use are traditionalist because they emphasize low-level cognitive processes (e.g., rote performance of a provided, step-by-step procedure; recall of given materials; focus on memorization and repetition to master skills). More research should explore the rationale for this result by gathering and analyzing teachers' own ideas about how and when technology should be integrated into classroom environments in addition to testing attitudes, beliefs, and pedagogical approaches.

Overall, results from this study suggest that pre-service teachers still have an incomplete idea of what *effective* technology integration practices and assessment looks like. That is, while pre-service teachers in this study could identify with the usefulness of technology as a tool to meet instructional and educator objectives, their beliefs and attitudes positively correlated with traditionalist approaches of technology integration. Thus, pre-service teachers may need sustained and iterative training in technology integration in order to move from traditionalist to constructivist approaches in its use.

Relationship between Prior Technology Experience, Attitudes/Beliefs, and Pedagogical Approaches

The second research question analyzed the relationships between pre-service teachers' prior technology usage and their agreement with student-centered technology pedagogical approaches. Results showed no significant correlations. The lack of relationship between technology experience and constructivist practices toward technology integration are inconsistent with previous research (Chai, 2010). Thus, a key question is why current pre-service teachers' approaches to technology integration may not be influenced by their personal expertise with technology.

One possibility is that the differences in technology expertise among pre-service teachers may have become less pronounced over time. Of the five categories of prior technology use utilized in this research (and drawn from previous studies), four commonly are used in academic and educational settings (word processing, spreadsheets, presentations (PowerPoint), and the Internet/World Wide Web). Thus, current pre-service teachers already may be more experienced with these technologies because modern university contexts require such proficiency. Technology experience with these "academic" tools may no longer be representative of pre-service teachers' comfort with technology in general. This possibility is supported by results concerning the fifth category of technology use targeted by this research: social media use. Social media usage is commonly not required in academic contexts and was the only technology studied in this research that is more limited to personal contexts. Self-rated social media use was the only category to be significantly and positively correlated with agreement that technology is valuable in meeting instructional objectives and that technology is important to instructional practices. Moreover, social media use (e.g., Twitter, Facebook, Instagram, etc.) was not positively correlated to confidence and experience in other forms of technology (i.e., word processing, spreadsheets, presentations, and internet/web). Thus, modern research may need to carefully separate “required” technology proficiency from experiences with “optional” and personal technologies (e.g., Facebook, Twitter, Instagram, Vine). It is possible that current pre-service teachers’ technology experiences are more sensitively measured by their non-academic experiences with technology tools and apps. Future research should address this possibility. However, it should be noted that the assumption that social media is not required in academic contexts is not always true. For example, the use of a classroom Twitter account to communicate activities and reminders to an outside audience has increased in popularity among technology-driven educators. It could be in the future that social media will become associated with educational technology (as word processing, spreadsheets, presentation software, and web-based activities are today) as educators realize the power social media presents in further enhancing instructional practices and overall student learning in the connected, digital age of tomorrow. Thus, future research may frequently need to explore and redefine the boundaries between required academic technologies and technologies that are personally-selected by upcoming generations of pre-service teachers.

Relationship between Technology Integration Course and Technology-based Pedagogy

The third research question addressed whether or not pre-service teachers who took a technology integration course would be more confident in using instructional technology, particularly in constructivist ways. Results indicated that participants who took a University of Utah introductory course in technology integration showed higher confidence in teaching with technology, integrating technology into future instruction, and using the web effectively than pre-service teachers who had not taken such a course. These results support past research that found a positive influence on pre-service teachers' attitudes and beliefs towards technology integration after taking a university introductory technology integration course (Anderson & Maninger, 2007; Bai & Ertmer, 2008; Funkhouser & Mouza, 2013). These results also suggest introductory technology integration courses may play an important role in influencing pre-service teachers' attitudes and beliefs towards technology. Current results suggest the importance of continuing to require a technology integration course in pre-service teacher education.

Results did not show an impact of taking a technology integration course on pre-service teachers' agreement with constructivist technology integration. This may be at least in part to the finding that all pre-service teachers' tended to agree strongly with the value of student-centered technology practices (averaging over 5 on a 6-point scale, where 6 = strongly agree). Thus, all pre-service may recognize the potential value of constructivist technology approaches, even if they do not feel confidence that they will implement such approaches. Another possibility is that pre-service teachers may broadly accept all examples of technology integration in classroom instruction as desirable, believing that technology is key to engaging students and creating 21st century classrooms. Thus, future research should explore methods that would provide sensitive and rich ways to document and assess pre-service teachers' understanding of technology integration.

Effect of Checklist on Evaluating Effective Technology Classroom Instruction

The fourth research question in this study examined what form of evaluation material (typical checklist, abstract checklist, or concrete checklist; see Appendix A) best facilitated analysis and application of digital materials for effective classroom instruction. In order to assess application, ratings of instructional vignettes were assessed before and after introduction to the checklist. Results showed no effect of test time, no effect of condition, and no interaction between time and condition. The absence of checklist effects as applied to instructional vignettes could reflect a lack of sufficient practice and feedback in using the checklists. The current study provided participants with a limited duration, single experience using the checklists and did not provide feedback on accuracy of checklist use. The brevity of the intervention as well as the lack of feedback may have limited pre-service teachers' understanding of and facility with the checklists. Another possibility is that, because participants were not explicitly instructed to use the checklists in the duration of the study, pre-service teachers may neglected to use them carefully during subsequent evaluation of general classroom descriptions. A final possibility is that pre-service teachers were not able to use the checklists accurately to evaluate the online lesson plans. This possibility was addressed by analyzing the accuracy of checklist usage in the abstract and concrete conditions.

Results showed a main effect of checklist type (abstract – principles vs. concrete – principles + cognitive examples) on participants' accuracy in evaluating online lesson plans (see Figure 1). Participants who used the concrete checklists were more accurate in evaluating the quality of technology integration found in online lessons. It could be that participants assigned to use the concrete checklist were more accurate at assessing online lesson plans because the presence of cognitive examples helped them accurately analyze varied examples of lesson plans. While the example were not explicitly related to the online lesson plans being evaluated, the examples nevertheless may have provided a better understanding of the abstract principles being assessed. This finding suggests that using realistic examples of student cognition for principles of technology integration may increase the utility of evaluation materials targeted toward pre-service and in-service teachers. Future research should continue to explore how and when these materials can be made more useful for educators.

Implications for Practice and Teacher Education

The results from this study have important potential implications for the education of pre-service and in-service teachers. The increased accuracy with which participants in the concrete checklist condition evaluated online lesson plans suggests that introductory technology courses and professional development on effective technology integration may benefit from the inclusion of cognitive examples of effective technology integration principles. The use of cognitive examples may be strengthened by explicit instruction on how to apply these examples across different digital contexts (e.g., how to identify "application of knowledge" in web-based collaborations, presentation (PowerPoint) tasks, and/or interactive online simulations). Educator training and professional development also should strengthen emphasis on known cognitive processes associated with different levels of learning (e.g., CI model: Kintsch, 1998). If educators are made aware of the benefits of teaching using high-level cognitive processes (e.g., analysis, evaluation, synthesis), then it may be more likely that they will recognize these processes in technology-related tasks and be more likely to implement constructivist approaches to technology integration.

The current research also suggests that there may be strong benefits associated with introducing pre-service teachers to a wide variety of technology programs, tools, and apps beyond those typically associated with the academic environment. Personal and creative use of non-academic technologies may predict the degree to which teachers are confident in using technology in innovative and non-traditional ways. Current association between social media use and technology values may indicate that technology-driven teachers are those who embrace technology in more than just the classroom environment.

Limitations and Suggestions for Future Research

One limitation of the present study is the small sample size of 30 participants. All participants were pre-service teachers at a single university in the mountain west; thus, the extent to which these results can be generalized to other pre-service teachers at other institutions is not known. In addition, because this study only analyzed one intervention over a short period of time (the duration of the study took between 1-2 hours), it may not be sensitive to the potential impact of different materials over longer intervention cycles. It may be necessary to track the influence of checklist materials over longer periods of time as well as with repeated interventions rather than just one experience.

Indeed, the limited exposure to materials in this study may be a particular concern. A study conducted by Funkhouser & Mouza (2013) examined the influence of an introductory course to technology in the classroom on teachers' beliefs about technology. Their results showed that use slightly changed to a mixed-method approach of teacher- and student-centered over a course of a semester. These results are important to remember when considering what is realistically achievable in an intervention module; it may not be feasible to significantly change pre-service teachers' pedagogical beliefs about learning with technology in one sitting as in this study. Furthermore, Ertmer (2005) advocates long-term interventions that focus on vicarious and/or personal experiences that add value to technology tools as the preferred method to shift teachers' technology pedagogical beliefs to student-centered approaches.

Future research on the topic of teachers' attitudes and beliefs towards technology could extend the current work by creating a training module on how to appropriately use the evaluation materials to identify effective technology integration. Specifically, this training module could show teachers how to correctly identify high-/low-level processes with the concrete checklist as a framework. For example, the training module could highlight one principle at a time in an online lesson plan while explicitly showing the participant how the lesson plan exemplifies or does not exemplify the principle. In addition, providing feedback through interactions between researcher and participant could aid teachers in accurately and effectively using the checklist to evaluate online materials. Implementing such a module as a long-term training that grows in difficulty and complexity may increase its potential value and impact. Future research should explore and address the necessary and sufficient conditions to improve technology integration and knowledge in training environments.

Conclusion

Overall, the current work demonstrated that past associations between educators' knowledge, beliefs, experiences, and pedagogy likely are not stable relationships and may be changing over time. Teachers who are "digital natives" appear to have different experiences and expectations about how, when, and why technology should be implemented in classrooms. Changing attitudes and skill sets among pre-service teachers necessitate the use of new materials and instructional methods that show the benefits of constructivist, student-centered pedagogical practices to technology integration. These new materials should have specific focus on high-level cognitive processes that optimize overall student learning. Current results show that pre-service teachers can accurately utilize materials that take this approach when evaluating online lesson plans. Continued research should explore training and support in helping pre-service teachers internalize and apply understanding of student learning to broader thinking about technology integration in classroom environments.

References

Albee, J. J. (2003). A study of pre-service elementary teachers' technology skill preparedness and examples of how it can be increased. *Journal of Technology and Teacher Education,* (11)1, 53-71.

Anderson, S. E., Groulx, J. G., & Maninger, R. M. (2011). Relationships among pre-service teachers' technology related abilities, beliefs, and intentions to use technology in their future classrooms. *Journal of Educational Computing Research,* 45(3), 321-338.

Anderson, S. E. & Maninger, R. M. (2007). Pre-service teachers' abilities, beliefs, and intentions regarding technology integration. *Journal of Educational Computing Research,* 37(2), 151-172.

Bai, H. & Ertmer, P. A. (2008). Teacher educators' beliefs and technology uses as predictors of pre-service teachers' beliefs and technology attitudes. *Journal of Technology and Teacher Education,* 16(1), 93-112.

Butcher, K. R., & Kintsch, W. (2012). Text comprehension and discourse processing. In A. F. Healy & R. W. Proctor (Eds.), *Handbook of Psychology, Volume Four: Experimental Psychology (2nd. Ed.)*. Hoboken, NJ: Wiley.

Butcher, K.R., Leary, H., Foster, J., & Devaul, H. (2014). Facilitating teachers' thinking about pedagogy and technology with an online curriculum planning tool. *Journal of Technology and Teacher Education, 22(4),* 423-447.

Chai, C. S. (2010). The relationship among Singaporean pre-service teachers' ICT competencies, pedagogical beliefs, and their beliefs on the espoused use of ICT. *The Asia-Pacific Education Research,* 19(3), 387-400.

Chi, M. T. H. (2009). Active-constructive-interactive: A conceptual framework for differentiating learning activities. *Topics in Cognitive Science, 1,* 73-105.

Chen, C. (2008). Why do teachers not practice what they believe regarding technology integration. *Journal of Educational Research,* 102(1), 65-75.

Culp, K. M., Honey, M., & Mandinach, E. (2005). A retrospective on twenty years of education technology policy. *Journal of Educational Computing Research, 32(3),* 279-307.

Ertmer, P. A. (2005). Teacher pedagogical beliefs: The final frontier in our quest for technology integration. *Educational Technology Research and Development,* 53(4), 25-39.

Ertmer, P. A. & Ottenbreit-Leftwich, A. T. (2010). Teacher technology change: how knowledge, confidence, beliefs, and culture intersect. *Journal of Research on Technology in Education,* 42(3), 255-284.

Ertmer, P. A. & Ottenbreit-Leftwich, A. T. (2013). Removing obstacles to the pedagogical changes required by Jonassen's vision of authentic technology-enabled learning. *Computers & Education,* 64, 175-182.

Ertmer, P. A., Ottenbreit-Leftwich, A. T., Sadik, O., Sendurur, E., & Sendurur, P. (2012). Teacher beliefs and technology integration practices: a critical relationship. *Computers & Education,* 59, 423-435.

Franklin, C. (2007). Factors that influence elementary teachers use of computers. *Journal of Technology and Teacher Education,* 15(2), 267-293.

Funkhouser, B. J. & Mouza, C. (2013). Drawing on technology: an investigation of pre-service teacher beliefs in the context of an introductory educational technology course. *Computers & Education,* 62, 271-285.

Hanson, K., & Carlson, B. (2005). Effective access: Teachers' use of digital resources in STEM teaching. Retrieved March 2, 2015 from http://www2.edc.org/gdi/publications_SR/EffectiveAccessReport.pdf

Hermans, R., Tondeur, J. B., & Vackle, M. (2008). The impact of primary school teachers' educational beliefs on the classroom use of computers. *Computers & Education,* 51, 1499-1509.

Judson, E. (2006). How teachers integrate technology and their beliefs about learning: is there a connection. *Journal of Technology and Teacher Education,* 14(3), 581-597.

Kim, C., Kim, M., Lee, C., Spector, M., & DeMeester, K. (2013). Teacher beliefs and technology integration. *Teacher and Teacher Education,* 29, 76-85.

Kintsch, W. (1988). The role of knowledge in discourse comprehension: A construction-integration model. *Psychological Review, 95*(2), 163-182.

Koehler, M. J., Mishra, P., & Yahya, K. (2007). Tracing the development of teacher knowledge in a design seminar: integrating content, pedagogy and technology. *Computers & Education,* 49, 740-762.

Liu, S. (2011). Factors related to pedagogical beliefs of teachers and technology integration. *Computers & Education,* 56, 1012-1022.

Ma, W., Andersson, R., & Streith, K. (2005). Examining user acceptance of computer technology: an empirical study of student teachers. *Journal of Computer Assisted Learning,* 21, 387-395.

Mitchem, K., Wells, D. L., & Wells, J. G. (2003). Effective integration of instructional technologies (IT): Evaluating professional development and instructional change. *Journal of Technology and Teacher Education, 11(3),* 397-414.

National Research Council. (2007). Taking science to school: Learning and teaching science in grades K-8. In R. A. Duschl, H. A. Schweingruber, & A. W. Shouse (Eds.). Washington, DC: The National Academies Press.

National Research Council. (2011). Successful K-12 STEM education: Identifying effective approaches in science, technology, engineering, and mathematics. Washington, DC: National Academies Press.

National Research Council. (2012). A framework for K-12 science education: Practices, crosscutting concepts, and core ideas. Washington, DC: National Academies Press.

National Science Board. (2010). Preparing the next generation of STEM innovators: Identifying and developing our nation's human capital. Arlington, VA: National Science Board.

Niederhauser, D. S. & Perkmen, S. (2007). Validation of the intrapersonal technology integration scale: assessing the influence of intrapersonal factors that influence technology integration. *Computers in the Schools,* 25(1-2), 98-111.

Ottenbreit-Leftwich, A. T., Glazewski, K. D., Newby, T. J., & Ertmer, P. A. (2010). Teacher value beliefs associated with using technology: addressing professional and student needs. *Computers & Education,* 55, 1321-1335.

Palak, D. & Walls, R. T. (2009). Teachers' beliefs and technology practices: a mixed-methods approach. *Journal of Research on Technology in Education,* 41(4), 417-441.

Pew Research Center (2010). "*Millennials: A Portrait of Generation Next.*" Retrieved from: http://pewsocialtrends.org/assets/pdf/millennials-confident-connected-open-to-change.pdf

Project Tomorrow. (2012). Personalizing the classroom experience: Teachers, librarians, and administrators connect the dots with digital learning *Speak Up 2011*. Irvine, CA: Project Tomorrow. Retrieved March 2, 2015 from http://www.tomorrow.org/speakup/pdfs/SU11_PersonalizedLearning_Educators.pdf.

Tondeur, J., Hermans, R., van Braak, J., & Valcke, M. (2008). Exploring the link between teachers' educational belief profiles and different types of computer use in the classroom. *Computers in Human Behavior,* 24, 2541-2553.

Valanides, N., & Angeli, C. (2008). Professional development for computer-enhanced learning: a case study with science teachers. *Research in Science & Technological Education, 26*(1), 3-12.

Vannatta, R. A. & Fordham, N. (2004). Teacher dispositions as predictors of classroom technology use. *Journal of Research on Technology in Education,* 36(3), 253-271.

Appendix A

1. Typical Checklist

Teacher Checklist for Evaluating Technology-Based Instruction

- ☐ Are students actively working with materials/technology/websites?
- ☐ Does the technology provide meaningful opportunities for learning?
- ☐ Are students using the technology to advance their thinking or knowledge?
- ☐ Will the technology be fun and engaging for students?

2. Abstract Checklist

Teacher Checklist for Evaluating Technology-Based Instruction

Generative Methods

Does the technology-based activity require students to create their own procedures or materials to solve a problem or answer a question? Look for:

- ☐ Creative problem-solving or coming up with new ideas
- ☐ Working on authentic questions (to which an answer is not known)
- ☐ Generating new materials, such as new representations or personalized content

Is the technology-based activity teacher-centered or student-centered? Look for:

Teacher-Centered

- ☐ Teacher presents or provides technology content to students
- ☐ Teacher manipulates the technology and/or directs interactions with the technology

Student-Centered

- ☐ Students find technology content or determine what/how to use digital information
- ☐ Students directly manipulate technology and/or determine how to interact with the technology

Quality of Cognitive Processing

Does the technology-based activity require only low-level cognitive processes? Look for:

- ☐ Memorization
- ☐ Repetition
- ☐ Rehearsal of provided information
- ☐ Recall of given materials
- ☐ Rote performance of a provided, step-by-step procedure

Does the technology-based activity require students to engage in high-level cognitive processes? Look for:

- ☐ Application of knowledge
- ☐ Inferences made about given content
- ☐ Integration of new and prior knowledge
- ☐ Evaluation of data or evidence
- ☐ Explanations that go beyond provided materials
- ☐ Focus on understanding concepts or ideas

3. Concrete Checklist

Teacher Checklist for Evaluating Technology-Based Instruction

Generative Methods

Does the technology-based activity require students to create their own procedures or materials to solve a problem or answer a question? Look for:

- ☐ Creative problem-solving or coming up with new ideas
  - *Examples: Develop a new solution, brainstorm ideas; provide unique examples*
- ☐ Working on authentic questions (to which an answer is not known)
  - *Examples: Where to put a new community center; design a cell phone for seniors*
- ☐ Generating new materials, such as new representations or personalized content
  - *Examples: Make a video; draw a diagram; create a website*

Is the technology-based activity teacher-centered or student-centered? Look for:

Teacher-Centered

- ☐ Teacher presents or provides technology content to students
  - *Examples: Show videos; play animations; present pictures*
- ☐ Teacher manipulates the technology and/or directs interactions with the technology
  - *Examples: Front-of-class display of a teachers' computer; students follow teacher-created process in a computer lab*

Student-Centered

- ☐ Students find technology content or determine what/how to use digital information
  - *Examples: Conduct web research; use online data to answer science questions*
- ☐ Students directly manipulate technology and/or determine how to interact with the technology
  - *Examples: Weather simulator where students manipulate environmental variables and observe results; students use infographic tool to create an informative poster*

Quality of Cognitive Processing

Does the technology-based activity require only low-level cognitive processes? Look for:

- ☐ Memorization
  - *Example: Instructional content tells students four key facts about the igneous rocks. Any questions focus on provided content (e.g., "What are four key facts about igneous rocks?")*
- ☐ Repetition
  - *Example: Students repeatedly practice one-digit addition problems that look similar (i.e., have the format: 2 + 2 = ____).*

- ☐ Rehearsal of provided information
    - *Example: Students are asked to repeat phases of the water cycle as presented by the instruction.*
- ☐ Recall of given materials
    - *Example: Students are asked to select three ways to tell if something is alive: all three correct choices were discussed in a previously-viewed video.*
- ☐ Rote performance of a provided, step-by-step procedure
    - *Example: In a weather simulator, students are told to set temperature to 60 degrees, humidity to 80%, and poleward temperature to 85% and record the results.*

Does the technology-based activity require students to engage in high-level cognitive processes? Look for:

- ☐ Application of knowledge
    - *Example: Students are told to use their knowledge of the heart and circulatory system to create an android with a more efficient circulatory system.*
- ☐ Inferences made about given content
    - *Example: Students are asked to explain why indigenous peoples from the northwest had homes made of wood whereas indigenous peoples from the southwest had homes made of clay (following instruction that presented both of these facts about people and shelter).*
- ☐ Integration of new and prior knowledge
    - *Example: Students are asked to think about how all the forms of precipitation that they can name are present in the water cycle.*
- ☐ Evaluation of data or evidence
    - *Example: Students use historical weather data to predict temperatures during their winter break.*
- ☐ Explanations that go beyond provided materials
    - *Example: Following a presentation that describes what germs are and how to wash your hands, students are asked to explain why hand sanitizer (which is very popular in schools!) helps to keep them healthy.*
- ☐ Focus on understanding concepts or ideas
    - *Example: Students are asked to compare and contrast the water cycle and the rock cycle, using examples from each one to explain what the word "cycle" means. (Previous instruction covered each cycle, but didn't explain what a cycle was.)*

Appendix B

High Constructivist Technology Use Vignette

Mrs. Kelly's 4th grade class is exploring how weather can be observed, measured, and recorded to make predictions about likely weather patterns. Mrs. Kelly creates four graphs of students to track weather patterns in different regions of the United States: the Pacific Northwest, Midwest, South, and North East regions. She explains that each group of students will work together to analyze their region's daily weather patterns with the goal of using the data collected to determine likely weather patterns of their region. Each day, Mrs. Kelly gives her class ample time to access a national weather forecasting site to determine daily precipitation rates, wind speeds, extreme storms, and temperatures. Each group then records the data collected in a Google Doc spreadsheet that is linked to the class website. Over the course of the weather unit, each group posts on Mrs. Kelly's 4th Grade Weather in the United States Blog each week to share any exciting weather happenings and to predict upcoming weather patterns based on past weather events in the area. At the end of the unit, each group uses the data collected to create a digital graph of one type of weather pattern (e.g., graph of the precipitation rates or of temperature changes). These graphs are then posted on the blog with an audio recording explaining how the students gathered the data and what types of predictions they can conclude from the graphs.

Medium Constructivist Technology Use Vignette

Mrs. Smith's 4th grade class is learning about weathering and erosion. Mrs. Smith thinks the best way for students to understand this example is to see real-life examples of weathering and erosion from nature – she believes strongly that students need to be able to connect ideas from her classroom to the world they see around them. She finds a large set of high-quality, detailed pictures using a Google Image search – the set of images that she selects depict rocks that have been weather down or eroded in various ways. In class, Mrs. Smith projects these images one at a time on her interactive smartboard for students to see. As each image is shown, Mrs. Smith uses the interactive smartboard tools to circle portions of the rock to show the class where the rock has been eroded or weathered down. As she works with each image, she explains to students the specific characteristics they can see that differentiate weathering and erosion as exemplified by the images. She also calls on students frequently to answer her questions about what they see in the image in order to keep students engaged. Once Mrs. Smith feels confident that the students understand the difference between weathering and erosion, she assesses their understanding by projecting similar but new images of rocks and rock landscapes and asks students to raise a hand if they think weathering has taken place (or keep their hand down if they think erosion has taken place). After providing students with enough think time, she asks for a student to volunteer an answer and explanation using details from the image and to explain their ideas. As they explain their ideas, Mrs. Smith circles the relevant features that they mention using the interactive whiteboard at the front of her class.

Low Constructivist Technology Use Vignette

Mr. Thompson's 5th grade class is learning about the three branches of government—legislative, executive, and judicial—and the responsibilities of each. Mr. Thompson created his own PowerPoint presentation to show the class, including short video clips describing how bills are made, images and diagrams depicting the checks and balances between the three branches, and a link to an interactive website that tests one's understanding on the responsibilities of the three branches of government through multiple choice questions. Mr. Thompson begins the lesson by projecting the presentation on the board for the class to see. Thompson goes through the slides with important information and shows the class the short student-friendly videos. After each video, he asks the class key questions that test what they remember from each video about the branch of government that was explained. At the end of the presentation, Mr. Thompson calls on students to volunteer one new and surprising fact that they learned. As a summative assessment, Mr. Thompson takes the class to the computer lab where they individually complete the online quiz that includes questions about definitions and facts for each branch of government. Mr. Thompson records each student's score as they complete the quiz so that he can decide on whether or not additional instruction will be necessary on this topic.

Appendix C

Vignette Evaluation Items (12 total)

Overall, this lesson:
Is a good example of how to integrate technology into the classroom
Uses the selected technologies to their maximum advantage
Shows an optimal example of how technology should be integrated into a classroom
This teacher selected:
Good examples of *authentic* online or digital materials
Good examples of *interactive* online or digital materials
Online or digital materials that go beyond what traditional materials could provide
This teacher created:
Meaningful opportunities for student learning overall
Opportunities for factual learning
Opportunities for learning broad concepts or ideas
Opportunities for students to take part in constructing their own knowledge
Opportunities for students to analyze or apply their knowledge
Opportunities for students to remember important facts or ideas
Opportunities for students to interact with technology in meaningful ways
Students would be:
Engaged in deep learning processes during the lesson
Engaged in 21st century learning skills during the lesson
Using technology in ways that are central to their learning

Appendix D

Survey items as compiled into technology attitudes/beliefs and pedagogical belief categories.

1. Technology valuable in meeting learning objectives

   Survey items:
   - I believe that using technology in the class activities will be time well spent.
   - I believe that students will be more motivated when they use computers for assignments.
   - I believe that when students use technology, they will create products that show high levels of learning.
   - I believe using online collaboration and communication tools (e.g., a Wiki) enhances student learning and cooperation skills.

2. Technology important to instructional practices

   Survey items:

   - Using instructional technology in the classroom will increase my productivity as a teacher.
   - Using instructional technology in the classroom will make it easier for me to teach.

3. Technology important to effectiveness as teacher

   Survey items:

   - Using instructional technology in the classroom will increase my effectiveness as a teacher.
   - Using instructional technology in the classroom will make my teaching more satisfying.
   - Using instructional technology in the classroom will make my teaching more exciting.
   - Effectively using instructional technology in the classroom will increase my sense of accomplishment.
   - I believe that I will be a better educator when I use technology for my work.

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